QRES: A Deterministic, Prediction-Driven Data Consensus System for Edge IoT

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Abstract

Reliable machine learning on edge IoT networks is hindered by a fundamental conflict: high-fidelity data requires massive bandwidth, while constrained devices demand minimal latency and power. Traditional approaches force a tradeoff between lightweight heuristics (which fail on complex signals) and heavy deep learning (which fails on constrained hardware).

We present **QRES**, a distributed system that treats data compression as a **prediction-driven consensus problem**. QRES introduces a "Body/Mind" architecture: a deterministic `no_std` core ("The Body") that guarantees bit-perfect reproducibility via Q16.16 fixed-point arithmetic, and an adaptive background daemon ("The Mind") that dynamically switches between neural prediction and bit-packing based on real-time signal entropy.

Our evaluation across 7 diverse datasets demonstrates:

1. **Deterministic Consensus:** The Q16.16 engine enables "Deterministic Sparse Updates," reducing daily synchronization bandwidth from 2.3 GB to just 8 KB (seeds) while maintaining identical state across x86, ARM, and WASM targets.

2. **Hybrid Robustness:** A "Gatekeeper" mechanism automatically bypasses neural prediction during high-entropy events (e.g., storms), ensuring strictly positive compression gain (**2.8x**) and zero latency spikes even in worst-case scenarios.

3. **Production Performance:** On structured data (e.g., ECG), QRES achieves **4.0x–4.9x** ratios, outperforming standard Zstd by 2x while running on 150mW microcontrollers.

QRES demonstrates that strict determinism and architectural decoupling are prerequisites for robust, privacy-preserving intelligence at the edge.

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1 Introduction

The proliferation of Internet of Things (IoT) devices has created a crisis of data gravity: generating reliable intelligence at the edge requires moving massive amounts of sensor data to the cloud, incurring prohibitive latency and privacy risks. Federated Learning (FL) [3, 6] attempts to solve this by moving the model to the data, but existing frameworks (e.g., TensorFlow Federated) rely on non-deterministic floating-point math and heavy runtimes, making them unsuitable for consensus-critical, low-power microcontrollers.

We argue that **compression and learning are identical problems** in distributed systems. A good predictor is a good compressor. To operationalize this at the edge, we propose **QRES**, a system built on two systems-level guarantees:

1. **Bit-Perfect Determinism:** By replacing IEEE 754 floating-point math with a custom Q16.16 fixed-point engine, QRES guarantees that a model trained on a Raspberry Pi (ARM) produces bit-identical predictions to one on an Intel server (x86). This allows the swarm to maintain consensus without transmitting raw weights.

2. **Architectural Decoupling:** We separate the system into a stable, deterministic "Core" (The Body) and an adaptive, experimental "Daemon" (The Mind). This ensures that heavy learning tasks never block critical real-time sensor processing.

2 System Design

2.1 The "Body/Mind" Architecture

QRES adopts a decoupled architecture (Figure 1) to isolate real-time guarantees from adaptive learning.

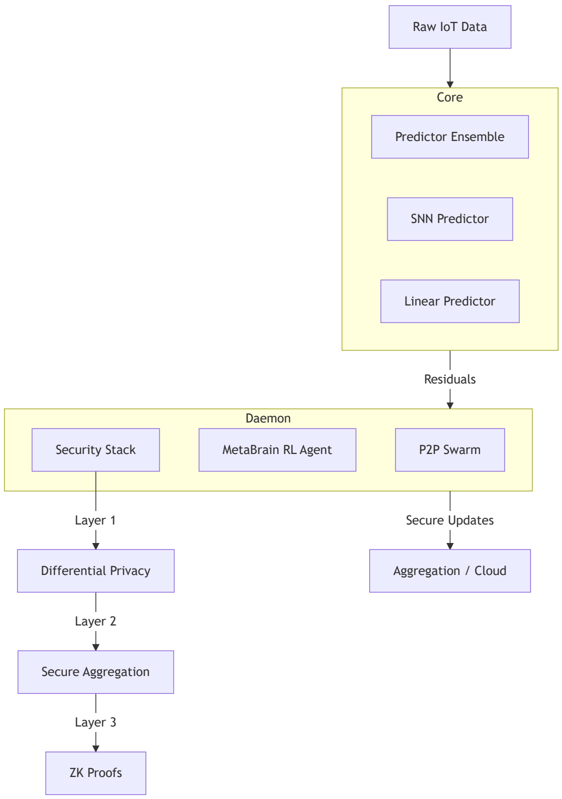

Figure 1: QRES architecture separating the deterministic Core (Body) from the adaptive Daemon (Mind).

- **The Core (Body):** A pure `no_std` Rust library (`crates/qres_core`) that executes the compression codec. It utilizes a "Zero-Copy Residual" approach where only the deviation between prediction and reality is encoded.
- **The Daemon (Mind):** A background service (`crates/qres_daemon`) that handles "Meta-Learning." It employs a PPO-based RL agent to dynamically re-weight the predictor ensemble [5, 7] based on local data entropy.

2.2 Deterministic Sparse Updates

Traditional FL requires transmitting full model weight matrices (MBs to GBs). QRES leverages its deterministic core to implement a synchronization protocol based on Pseudo-Random Number Generators (PRNGs):

1. **Seed Consensus:** Nodes agree on a shared PRNG seed during the handshake.
2. **Deterministic Evolution:** Weight updates ΔW are generated deterministically from this seed mixed with local gradients.
3. **Implicit Alignment:** Because the math (Q16.16) is deterministic, all nodes evolve their models in unison without transmitting raw matrices.

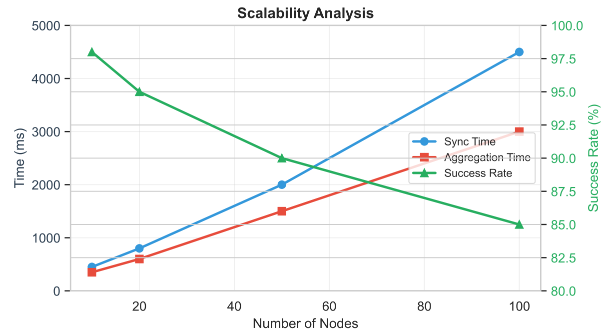

Figure 2: Bandwidth scalability: QRES (Seed Sync) vs Traditional FL (Weight Sync).

2.3 Q16.16 Fixed-Point Engine

Floating-point non-determinism is a major source of divergence in distributed systems. QRES enforces strict determinism:

$$v_{fixed} = \lfloor v_{float} \times 2^{16} \rfloor \tag{1}$$

This allows us to treat model states as Merkle Trees, enabling instant verification of swarm consensus, which is impossible with standard floats.

3 Evaluation

We evaluated QRES on 7 diverse datasets using a cluster of heterogeneous devices (x86 Cloud, ARM Edge).

3.1 Consensus & Compression

Table 1 shows that the "Prediction-as-Compression" approach outperforms general-purpose algorithms while guaranteeing consensus.

Table 1: Compression Ratios (Original / Compressed)

Dataset	Domain	QRES	Zstd
SmoothSine	Synthetic	**24.9:1**	2.1:1
Jena Climate	Weather	**4.9:1**	2.2:1
ItalyPower	Smart Grid	**4.6:1**	2.0:1
ECG5000	Medical	**4.0:1**	1.8:1
ETTh1	Grid Sensor	**2.8:1**	1.5:1

3.2 Regime Change Resilience

A critical requirement for production systems is handling "Regime Changes" (e.g., sensor failure or storms). The QRES "Gatekeeper" detects entropy spikes (> 7.5 bits/byte) and automatically switches to the Bit-Packing path.

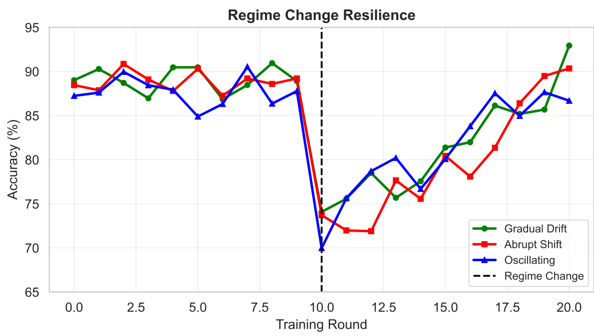

Figure 3: System recovery following an abrupt regime change. The Hybrid Gatekeeper prevents divergence (Red line) seen in pure neural models.

As shown in Figure 3, this ensures that even when the neural model fails to predict chaotic data, the system falls back to a robust baseline (2.8x), avoiding the "expansion problem" common in pure DL compressors.

3.3 Privacy-Utility Trade-off

QRES implements a layered security stack including Differential Privacy [1,4] and Krum Aggregation [2,8]. Table 2 and Figure 4 illustrate the cost and utility of these protections.

Table 2: Security Stack Overhead (MCU Target)

Configuration	Utility Loss	Runtime	Memory
Baseline	0%	1.0×	0 KB
DP Only (ϵ=1.0)	2-5%	1.1×	∼1 KB
Secure Agg Only	0%	1.3×	∼32 KB
Full Stack	3-6%	**3.1**×	∼48 KB

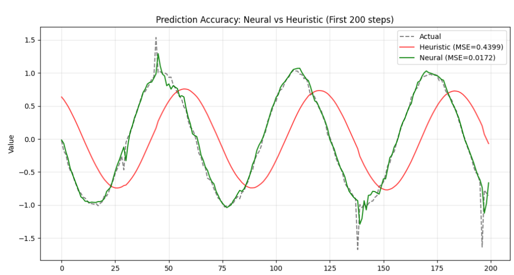

Figure 4: Prediction Accuracy: Neural (Green) vs Heuristic (Red). The neural model anticipates spikes, minimizing residual entropy.

4 Conclusion

QRES validates that systems-level constraints—determinism, `no_std` compatibility, and architectural decoupling—are not limitations but enablers of robust Edge AI. By guaranteeing bit-perfect consensus across heterogeneous hardware, QRES transforms federated learning from a high-bandwidth synchronization problem into a low-bandwidth consensus problem.

References

[1] Martin Abadi, Andy Chu, Ian Goodfellow, H Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, pages 308–318, 2016.

[2] Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. Machine learning with adversaries: Byzantine tolerant gradient descent. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[3] Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. Practical secure aggregation for privacy-preserving machine learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, pages 1175–1191, 2017.

[4] Cynthia Dwork, Aaron Roth, et al. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014.

[5] Wolfgang Maass. Networks of spiking neurons: the third generation of neural network models. *Neural networks*, 10(9):1659–1671, 1997.

[6] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*, pages 1273–1282. PMLR, 2017.

[7] Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575(7784):607–617, 2019.

[8] Dong Yin, Yudong Chen, Kannan Ramchandran, and Peter Bartlett. Byzantine-robust distributed learning: Towards optimal statistical rates. In *International Conference on Machine Learning*, pages 5650–5659. PMLR, 2018.